Lecture Notes, Math Workshop 2026

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Contents

1 Basic Mathematics and Notation

1.1 About this Document

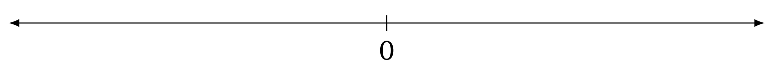
What is the purpose of this course? These notes constitute lectures about basic math, linear algebra, and calculus for incoming Ph.D. students to Ohio State's Political Science graduate program. The course runs for two weeks and no mathematical background behind high school algebra is assumed. That said, prior math training among incoming students tends to vary widely. While some students have taken extensive relevant coursework in college, others have not picked up a mathematics textbook since their senior year in high school. The course is designed with this variation in training in mind. Students already exposed to the following concepts will benefit from a review — there is always more to learn – and the material covered is intentionally cursory so that those without prior experience are not overwhelmed. Either way, when presented with new mathematical concepts in subsequent statistics or formal theory courses, students will have a set of notes and enough prior exposure to help them figure out what equations represent and how to solve them.

Although not all of the following material is drawn from Moore and Siegel (2013), the overlap is sizeable and intentional. Many thanks to the authors. Some of the examples are also drawn from Jason Morgan's lectures (2015) with additions by Drew Rosenberg and Daniel Kent. This document was originally compiled by Daniel Kent.

1.2 Variables and Constants

- **Variable**: A concept or a measure that takes different values in a given set; e.g., GDP, polity score, party identification, etc.
- **Constant**: A concept or a measure that has a single value for a given set.

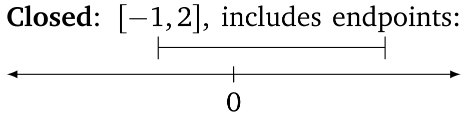
1.3 Number Types and Notation



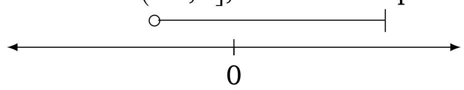
- **Real numbers**: ($\mathbb{R}$), can be placed anywhere on the line
- **Natural numbers**: ($\mathbb{N}$), positive with no decimal
- **Integer**: ($\mathbb{Z}$), non-decimal, both positive and negative
- **Rational number**: can be expressed as a ratio or fraction
- **Irrational number**: cannot be expressed as a fraction: π, e
- $\mathbb{R}^k$ is a k-dimensional space

1.4 Intervals

- **Open**: $(-1, 2)$, does not include endpoints, so greater than -1 and less than 2.
- **Closed**: $[-1, 2]$, includes endpoints: greater than or equal to -1 and less than or equal to 2.



- **Half-closed**: $(-1, 2]$, includes endpoints: greater than or equal to -1 and less than or equal to



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1.5 Levels of measurement

- **Nominal**: No mathematical relationship among the values
  - E.g., race, gender, country, party, religion
- **Ordinal**: Ranking (or ordered); cannot do arithmetic because the distance between values is not equal.
  - E.g., K-12, age cohort, ideology (left-to-right)
- **Interval**: Fixed differences, but zero is arbitrary. Percentage change is not well defined.
  - E.g., temperature, date/time
- **Ratio**: Fixed differences with a true zero
  - E.g., age, length, income, votes

1.6 Sets

- A **set**, $S = \{\}$, is a collection of elements.
  - Order does not matter: $S = \{x, y, z\} = \{z, y, x\}$
    - $x \in S$, meaning x is an element of S
    - $a \notin S$, meaning a is not an element of S
- Other examples:
  - $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$ or $\{x \mid x \text{ is an integer}\}$
  - $T = \{x \mid 8 \geqslant x \geqslant 7\}$ or $T = [7, 8]$
- Set notation:
  - $S = \{x, y, z\}$, $T = \{a, b, x\}$, $N = \{a, b\}$
    - Intersection: $T \cap S = \{x\}$
      - $N \cap S = \varnothing$: “empty set”
    - Union: $N \cup S = \{a, b, x, y, z\}$
    - Complement (not in the set): $(T \cap S)^c = \{a, b, y, z\}$
  - $\mathbb{U}$: universal set, all possible values
  - $A = \{x, y\}$
    - $A \subset S$: we say “A is a subset of S”
    - If $\not\subset$, then not a subset of. All subsets of A: $\{x\}$, $\{y\}$, $\{\varnothing\}$, $\{x, y\}$

- The null set, $\{\varnothing\}$, is a subset of all sets. Can you prove this?

- Transitivity:
  - If $Z \in Q$ and $Q \in R$, then $Z \in R$
- Disjoint:
  - No elements in common; more formally, two sets are disjoint if the intersection of the sets is the null set: $N \cap S = \{\varnothing\}$

1.7 Functions

- $f(x) = y$; read this as "y is some function of x"
- **Constant function**: a function whose outcome is the same no matter the input. If $f(x) = c$, then no matter x the output is c.
- **Polynomial function**: $y = a + bx + cx^2$
- **Rational function**: can be defined by a fraction (ratio):

$$f(x, y) = \frac{a + bx^2}{1 + y}$$

1.8 Independent and Dependent Variables

Let $y = f(x)$, where y is the outcome and x the input. (Note that $f(x)$ is a function; see the next section.)

- **Independent variable**: the input; e.g., x
- **Dependent variable**: outcome; e.g., y

In a linear model, i.e. $y = \alpha + \beta x$, the dependent variable is y and the independent variable is x.

- **Endogenous**: "comes from within"; y is called the endogenous variable.
- **Exogenous**: "comes from without"; x is called the exogenous variable.

Important: this implies *nothing* about causation; be careful with this language and the temptation to draw causal conclusions from a mathematical relationship. (See the works of David A. Freedman, Paul Holland, Donald Rubin.)

1.9 Inequalities and Absolute Values

$$|x| = \sqrt{x^2} = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Useful properties:

- $|m| + |n| \geqslant |m + n|$
- $|m| \times |n| = |m \times n|$

1.10 Exponent Rules

- $x^n = x \cdot x \cdot x \cdot x \cdot \ldots \cdot x$ n times; $2^2 = 2 \cdot 2 = 4$
- $x^0 = 1$ (Why? Show using the rules of division of exponents; but we typically say it's undefined when $x = 0$)
- $x^m \cdot x^n = x^{m+n}$
- $\frac{x^m}{x^n} = x^{m-n}$
- $\frac{1}{x^m} = x^{-m}$
- $x^1 = x$
- $x^{\frac{1}{m}} = \sqrt[m]{x}$
- $(x^m)^n = x^{m \cdot n}$
- $x^m \cdot y^m = (xy)^m$

But:

- $x^m + y^n \neq (x + y)^{m+n}$

1.11 Commutative, Associative, and Distributive Laws

- **Associative property**: rewriting the parentheses does not change the result
  - $(a + b) + c = a + (b + c)$
  - $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- **Commutative property**: order of operands does not change the result
  - $a + b = b + a$
  - $a \cdot b = b \cdot a$
- **Distributive property**:
  - $a(b + c) = ab + ac$

Also relevant:

- **Inverse property**:
  - **Additive Inverse**: For any x, there exists a $-x$ such that $-x + x = 0$.
    - Formally[1]: $\exists(-x)$ s.t. $-x + x = 0$
  - **Multiplicative Inverse** For any x, there exists a x^{-1} such that $x \cdot x^{-1} = 1$.
    - Formally: $\exists(x^{-1})$ s.t. $x^{-1} \cdot x = 1$
- **Identity property**:
  - $\exists(0)$ s.t. $x + 0 = x$
  - $\exists(1)$ s.t. $x \cdot 1 = x$
  - Commonly, we see: $\mathrm{I}(x) = x$

1.12 Summation Operator

- Consider a set $X = \{x_1, x_2, ..., x_n\}$
  - $\sum\limits_{i=1}^{n} x_i = x_1 + x_2 + ... + x_n$
    - "The sum of x_i, over the range from $i = 1$ through $i = n$."
- Let $Y = \{y_1, y_2, ..., y_n\}$
  - $\sum\limits_{i=1}^{n} x_i y_i = x_1 y_1 + x_2 y_2 + ... + x_n y_n$
    - Here, we sum the outcome of $x_i y_i$ n times.
- But, what if there is more than one summation operator and subscript?

$$\begin{aligned}\Sigma_{j=1}^{m}\Sigma_{i=1}^{n} x_i y_j &= (y_1 x_1 + y_1 x_2 + ... + y_1 x_n) \\ &\quad + (y_2 x_1 + y_2 x_2 + ... + y_2 x_n) \\ &\quad + ... + (y_m x_1 + y_m x_2 + ... + y_m x_n)\end{aligned}$$

  We have m sets of parentheses, with n x's in each parenthesis. We end up summing each combination of x_i and y_j.

- What if there is a constant, c? Drawing upon the *distributive property*:
  - $\sum\limits_{i=1}^{n} c x_i = c x_1 + c x_2 + ... + c x_n = c \sum\limits_{i=1}^{n} x_i$
- The *associative property* can also be applied to summation:
  - $\sum\limits_{i=1}^{n} (x_i + y_i) = (x_1 + y_1) + (x_2 + y_2) + ... + (x_n + y_n) = \sum\limits_{i=1}^{n} x_i + \sum\limits_{i=1}^{n} y_i$

[1] $\exists$ = 'there exists'

1.13 Product Operator

- $X = \{x_1, x_2, ..., x_n\}$ and $Y = \{y_1, y_2, ..., y_n\}$
- $\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdot ... \cdot x_n$
- $\prod_{i=1}^{n} (x_i + y_i) = (x_1 + y_1)(x_2 + y_2)...(x_n + y_n)$
    - We can't split $(x_i + y_i)$ like in summation. Instead, we multiply $(x_i + y_i)$ repeatedly.
- What about a constant?
    - $\prod_{i=1}^{n} cx_i = (cx_1)(cx_2) \cdot ... \cdot (cx_n) = c^n \prod_{i=1}^{n} x_i$

  We can move c to the front, but we have to exponentiate it to c^n.

1.14 Solving equations, inequalities, and for roots

- Solving an equation example:

$$\begin{aligned} 3x + 4y + 8 &= 0 \\ 4y &= -(3x + 8) \\ y &= \frac{-(3x + 8)}{4} \\ y &= -\frac{3}{4}x - 2 \end{aligned}$$

  We can also write this answer as:

$$\{(x, y) \in \mathbb{R} \mid y = -\frac{3}{4}x - 2\}$$

- Solving an inequality example:

$$\begin{aligned} -4y &> 2x + 12 \\ y &< -\frac{2x}{4} - \frac{12}{4} \\ y &< \frac{x}{2} - 3 \end{aligned}$$

  Note that dividing by a negative *flips the inequality*. Multiplying by a negative does the same.

- Solving for a quadratic

- Quadratic formula:

$$
\begin{aligned}
&ax^2 + bx + c \\
&x \in \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{aligned}
$$

  Example:

$$
\begin{aligned}
&1.4x^2 + 3.7x + 1.1 = 0 \\
&a = 1.4, b = 3.7, c = 1.1 \\
&x = \frac{-3.7 \pm \sqrt{3.7^2 - 4 \times 1.4 \times 1.1}}{2.8} \\
&x = -0.341 \text{ or } x = -2.301
\end{aligned}
$$

1.15 Logarithms

- If $y = a^x$, then we can rewrite as $\log_a y = x$
- Examples:
  - $\log_e e = 1$, because $e^1 = e$
  - Solving for x:

    If $8 = 2^x$, then $\log_2 8 = x = 3$

```
# To solve in R:
> log(8, base=2)    # defaults to base=exp(1)
```

  - What is e?

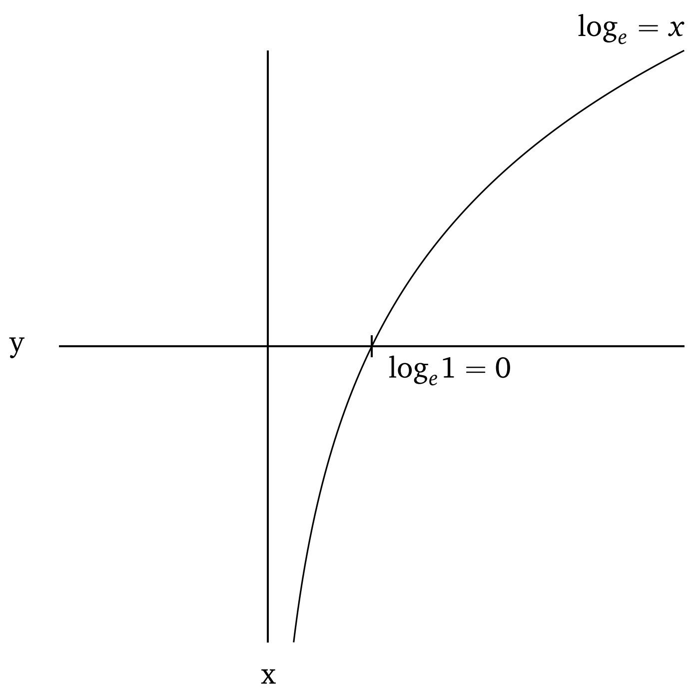
$$
e = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x
$$

```
# To plot this in R:
> x <- 1:1000
> y <- (1 + (1/x))^x
> plot(x, y, type="l", xlab="x", ylab="value")
```

- **Rules**
  - $\log(m \cdot n) = \log(m) + \log(n)$
  - $\log(\frac{m}{n}) = \log(m) - \log(n)$
  - $\log(b^a) = a \log b$
  - $\log_b(a) = (\log_b e)(\log_e a)$

- $\log_b(a) = \frac{1}{\log_a b}$

- **Visual**



Note: $\log_e(x) \equiv \ln(x)$, pronounced the 'natural logarithm'

```
# Approximately replicate this in R:
> x <- seq(0.1, 4, by=0.1)
> y <- log(x)
> plot(x, y, type="l", xlab="x", ylab="y", xlim=c(-1, 4),
+       lwd=2)
> abline(v=0, h=0)
> text(3, 1.35, labels=expression("log"["e"]*"=x"))
```

2 Introduction to Probability and Statistics

This module introduces the fundamental concepts of probability theory and statistics that form the basis of quantitative political methodology. In social science, we use probability models to represent uncertainty, make inferences about populations from samples, and model decision-making behavior.

2.1 Random Experiments, Sample Spaces, and Events

To build a mathematical theory of probability, we start with set theory concepts.

- **Random Experiment**: A process or trial that leads to one of several possible outcomes, where the exact outcome is uncertain before the experiment is run.
  - E.g., flipping a coin, rolling a die, or holding an election.
- **Sample Space** (S or Ω): The set of all possible outcomes of a random experiment.
  - E.g., for a single coin flip, $S = \{\text{Head}, \text{Tail}\}$.
  - E.g., for a roll of a six-sided die, $S = \{1, 2, 3, 4, 5, 6\}$.
- **Event** (A): Any subset of the sample space ($A \subset S$). An event is said to occur if the outcome of the experiment is an element of A.
  - E.g., rolling an even number on a die: $A = \{2, 4, 6\}$.
  - E.g., the universal event S (something happens) and the empty/impossible event $\varnothing$ (nothing happens).

2.2 Set-Theoretic Relations on Events

Since events are sets, we can define relations between events using standard set operators:

- **Union** ($A \cup B$): The event that A occurs, OR B occurs, OR both occur.
- **Intersection** ($A \cap B$): The event that BOTH A and B occur.
- **Complement** (A^c or $\bar{A}$): The event that A does NOT occur.
- **Mutually Exclusive (Disjoint)**: Two events A and B are mutually exclusive if they cannot occur at the same time, meaning $A \cap B = \varnothing$.

2.3 Factorials, Permutations, and Combinations

Most of our quantitative coursework is about modeling probabilities, where:

$$\text{probability} = \frac{\#\text{occurences}}{\#\text{possibilities}}$$

For both the numerator and denominator we are dealing with *counting* the number of relevant outcomes. Factorials, permutations, and combinations are foundational concepts when it comes to counting.

- Some useful illustrations/properties:
  - $0! = 1$
  - $x! = x \cdot (x-1) \cdot (x-2) \cdot \ldots \cdot 0!$
  - $2! = 2 \cdot 1$
  - $3! = 3 \cdot 2 \cdot 1$
  - $10! = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot \ldots \cdot 1$
  - If n objects in boxes of size k where *order matters*, then the number of *permutations* is:

    $$\frac{n!}{(n-k)!} = nPk$$

  - If order does not matter, then the number of *combinations* is:

    $$\frac{n!}{(n-k)!\,k!} = nCk = \binom{n}{k}$$

    - Pronounced: 'n choose k'

2.4 Axioms of Probability

A probability function $P(\cdot)$ maps any event $A \subset S$ to a real number $P(A)$ representing its likelihood. To be mathematically valid, $P(A)$ must satisfy Kolmogorov's three axioms:

1. **Non-negativity**: For any event A, $P(A) \geqslant 0$.
2. **Normalization**: The probability of the sample space (certain event) is 1, i.e., $P(S) = 1$.
3. **Additivity of Disjoint Events**: For any sequence of mutually exclusive events $A_1, A_2, \ldots$ (where $A_i \cap A_j = \varnothing$ for all $i \neq j$):

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

These axioms imply several useful rules:

- Complement Rule: $P(A^c) = 1 - P(A)$
- Probability of Empty Set: $P(\varnothing) = 0$
- Range of Probability: $0 \leqslant P(A) \leqslant 1$
- Monotonicity: If $A \subset B$, then $P(A) \leqslant P(B)$

- General Addition Rule (for any two events):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2.5 Conditional Probability and Independence

Often, we want to update the probability of an event A given that another event B has already occurred. This is called conditional probability.

- **Conditional Probability**: The probability of event A given event B (where $P(B) > 0$) is defined as:
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$
This implies the Multiplication Rule: $P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$.

- **Independence**: Two events A and B are independent if the occurrence of one does not affect the probability of the other:

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B)$$

Equivalently, A and B are independent if and only if:

$$P(A \cap B) = P(A)P(B)$$

2.6 Law of Total Probability and Bayes' Rule

- **Partition**: A collection of disjoint events $B_1, B_2, \ldots, B_k$ that cover the entire sample space, i.e., $B_i \cap B_j = \varnothing$ for $i \neq j$ and $\bigcup_{i=1}^{k} B_i = S$.

- **Law of Total Probability**: If $B_1, \ldots, B_k$ is a partition of S, then for any event A:

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i)$$

- **Bayes' Rule**: Allows us to invert conditional probabilities. If $B_1, \ldots, B_k$ is a partition of S and $P(A) > 0$, then:
$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)} = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{k} P(A \mid B_i)P(B_i)}$$
In political science, Bayes' Rule is the foundation of Bayesian statistical inference (updating prior beliefs $P(B_j)$ with new data/evidence A to get posterior beliefs $P(B_j \mid A)$) and game-theoretic models of incomplete information.

2.7 Random Variables and Probability Distributions

Instead of working with abstract outcomes (like "Heads" or "Tails"), we map outcomes to numbers using random variables.

- **Random Variable** (X): A function that maps each outcome in a sample space S to a real number ($X : S \to \mathbb{R}$).
  - E.g., let X be the number of heads in two coin flips. If outcomes are $\{TT, HT, TH, HH\}$, then $X(TT) = 0$, $X(HT) = 1$, $X(TH) = 1$, and $X(HH) = 2$.
- **Discrete Random Variable**: A random variable that can take on a countable number of distinct values (finite or countably infinite).
  - Managed via a **Probability Mass Function (PMF)** $f(x) = P(X = x)$, where:

$$\sum_x f(x) = 1 \quad \text{and} \quad f(x) \geqslant 0 \quad \text{for all } x$$

- **Continuous Random Variable**: A random variable that can take on any value in an interval or range of real numbers (uncountably infinite).
  - Managed via a **Probability Density Function (PDF)** $f(x)$, where $P(a \leqslant X \leqslant b) = \int_a^b f(x)dx$. Crucially:

$$\int_{-\infty}^{\infty} f(x)dx = 1 \quad \text{and} \quad f(x) \geqslant 0 \quad \text{for all } x$$

    Note that for a continuous random variable, the probability of taking any exact single value is zero: $P(X = x) = 0$.
- **Cumulative Distribution Function (CDF)** ($F(x)$): The probability that a random variable X is less than or equal to a value x:

$$F(x) = P(X \leqslant x)$$

  - For discrete variables: $F(x) = \sum_{t \leqslant x} f(t)$
  - For continuous variables: $F(x) = \int_{-\infty}^{x} f(t)dt$
  - Properties: $F(x)$ is non-decreasing, $\lim_{x\to-\infty} F(x) = 0$, and $\lim_{x\to\infty} F(x) = 1$. By the Fundamental Theorem of Calculus, for continuous variables, $F'(x) = f(x)$.

2.8 Expectation, Mean, and Variance

We summarize random variables using their theoretical moments, such as the mean (central tendency) and variance (dispersion).

- **Expected Value** ($E[X]$ or μ): The long-run average value of a random variable.

- For a discrete random variable:
$$E[X] = \sum_{x} x \cdot f(x)$$
- For a continuous random variable:
$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

- Expected Value of a Function $g(X)$:
  - Discrete: $E[g(X)] = \sum_x g(x)f(x)$
  - Continuous: $E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$
- Properties of Expectation (Linearity):
  - $E[c] = c$ for any constant c.
  - $E[cX] = cE[X]$ for any constant c.
  - $E[X + Y] = E[X] + E[Y]$ for any random variables X and Y (even if dependent).
- **Variance** ($Var(X)$ or σ^2): A measure of the spread of a random variable around its expected value:
$$Var(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$
- Properties of Variance:
  - $Var(c) = 0$ for any constant c.
  - $Var(cX) = c^2 Var(X)$ for any constant c.
  - If X and Y are independent: $Var(X + Y) = Var(X) + Var(Y)$.
- **Standard Deviation** (σ): The square root of the variance, expressed in the same units as the random variable: $\sigma = \sqrt{Var(X)}$.

3 Linear Algebra

Most, if not all, algebra learned throughout K-12 education deals with *scalar* algebra. Each variable only represents a single number. But, a variable can represent more than one element. Instead:

- *Scalar*: one element, x
- *Vector*: n elements
  - $\boldsymbol{x}$ includes the set of scalars $x_1, x_2, ..., x_i, ..., x_n$
- *Matrix*: $n \times m$ elements
  - $\boldsymbol{X}$ includes the set of scalars $x_{11}, ..., x_{1m}, x_{21}, ..., x_{2m}, ..., x_{n1}, ..., x_{nm}$

There are considerable gains to be made through this notation. Beyond a more efficient notation, we're able to easily carry out all sorts of useful manipulations. We'll work up to these benefits in this lecture through various concepts.

3.1 Systems of Equations

- *System of equations*: two or more equations with the same variables
  - To find a unique solution, we need as many equations as variables. E.g.,

$$\begin{aligned} 6x_1 - 3x_2 + 4x_3 &= -13 \\ 6x_1 &= -13 + 3x_2 - 4x_3 \\ x_1 &= -\frac{13}{6} + \frac{1}{2}x_2 - \frac{2}{3}x_3 \end{aligned}$$

  - We can use this single equation and solution to create a series of solutions, i.e. if $x_2 = 2$ and $x_3 = 3$, then $x_1 = -\frac{19}{6}$ and so on.

- Consider the following system of (linear) equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + ... + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + ... + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + ... + a_{mn}x_n &= b_m \end{aligned}$$

  Drawing on yesterday's lecture, we can also write this all as: $\sum_{j=1}^{n} a_{ij}x_j = b_i$ for each $i = 1, \dots, m$. Or, bringing matrix notation in, as: $\boldsymbol{Ax} = \boldsymbol{b}$, where $\boldsymbol{A}$ is an $m \times n$ matrix.

3.1.1 Solving a system of equations

Consider the following system of equations. Throughout r references row number. By setting all variables but one per equation, we are solving by *elimination*. We'll later introduce Gauss-Jordan elimination, which is the same process, but with an augmented matrix.

- First, the equations:

$$\begin{aligned} 1x_1 + 1x_2 &= 100{,}000 \\ 0.05x_1 + 0.09x_2 &= 7{,}800 \end{aligned}$$

- Add $-0.05\,r1$ to $r2$

$$\begin{aligned} 1x_1 + 1x_2 &= 100{,}000 \\ 0x_1 + 0.04x_2 &= 2{,}800 \end{aligned}$$

- $25\,r2$

$$\begin{aligned} 1x_1 + 1x_2 &= 100{,}000 \\ 0x_1 + 1x_2 &= 70{,}000 \end{aligned}$$

- Subtract $r2$ from $r1$

$$\begin{aligned} 1x_1 + 0x_2 &= 30{,}000 \\ 0x_1 + 1x_2 &= 70{,}000 \end{aligned}$$

- This gives us: $x_1 = 30{,}000$ and $x_2 = 70{,}000$

Now, let's try a lengthier example. Again, we're *not substituting* equations into each variable. We're eliminating all variables but one from each equation.

- First, the equations:

$$\begin{aligned} 1x_1 + 2x_2 + 3x_3 &= 6 \\ 2x_1 - 3x_2 + 2x_3 &= 14 \\ 3x_1 + 1x_2 - 1x_3 &= -2 \end{aligned}$$

- Subtract $2\,r1$ from $r2$ and subtract $3\,r1$ from $r3$

$$\begin{aligned} 1x_1 + 2x_2 + 3x_3 &= 6 \\ 0x_1 - 7x_2 - 4x_3 &= 2 \\ 0x_1 - 5x_2 - 10x_3 &= -20 \end{aligned}$$

- Multiply $r3$ by $-\frac{1}{5}$ and flip $r2$ and $r3$

$$\begin{aligned} 1x_1 + 2x_2 + 3x_3 &= 6 \\ 0x_1 + 1x_2 + 2x_3 &= 4 \\ 0x_1 - 7x_2 - 4x_3 &= 2 \end{aligned}$$

- Subtract $2\,r2$ from $r1$ and add $7\,r2$ to $r3$

$$\begin{aligned} 1x_1 + 0x_2 - 1x_3 &= -2 \\ 0x_1 + 1x_2 + 2x_3 &= 4 \\ 0x_1 + 0x_2 + 10x_3 &= 30 \end{aligned}$$

- Divide $r3$ by 10

$$\begin{aligned} 1x_1 + 0x_2 - 1x_3 &= -2 \\ 0x_1 + 1x_2 + 2x_3 &= 4 \\ 0x_1 + 0x_2 + 1x_3 &= 3 \end{aligned}$$

- Add $r3$ to $r1$ and subtract $2\,r3$ from $r2$

$$\begin{aligned} 1x_1 + 0x_2 - 0x_3 &= 1 \\ 0x_1 + 1x_2 + 0x_3 &= -2 \\ 0x_1 + 0x_2 + 1x_3 &= 3 \end{aligned}$$

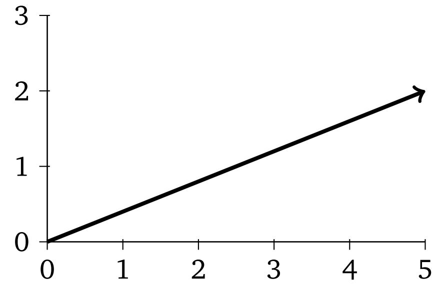
- $x_1 = 1, x_2 = -2, x_3 = 3$

Writing all of the $x's$ gets tedious though. So let's introduce vectors.

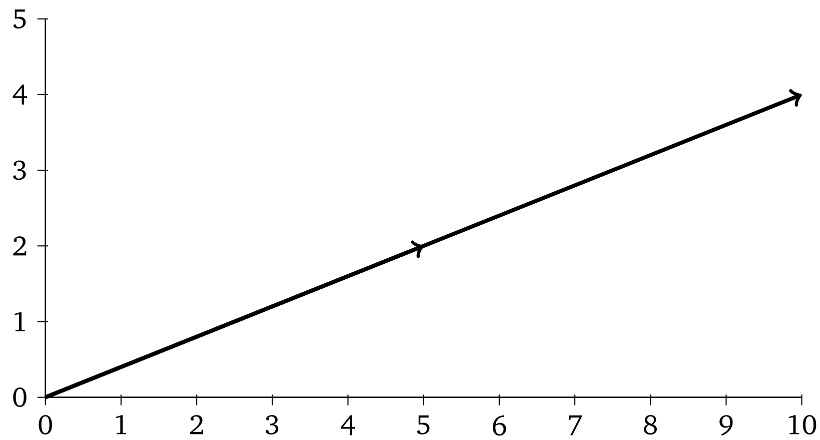
3.2 Vectors

- Examples:
  - $\boldsymbol{x} = [x_1, x_2, x_3, ..., x_n]$

- $\boldsymbol{x} = [5, 2]$
  - If, like in this case, there are two dimensions (number of components), then we can visually understand the vector as:



- If we multiply by a scalar: $a = 2$, $a\boldsymbol{x} = [10, 4]$



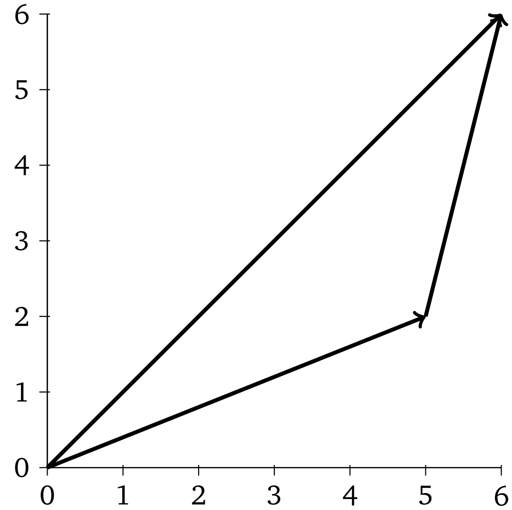
3.2.1 Vector length

- Also called the *norm*, length is not the same as *dimensions*. The formula is:

$$||\boldsymbol{x}|| = \sqrt{x_1^2 + x_2^2 + ... + x_n^2}$$

- Considering the visuals above, in two dimensions we are just using the Pythagorean theorem. But the formula is extendable to k-dimensions. For $\boldsymbol{x} = [5, 2]$, $||\boldsymbol{x}|| = \sqrt{25 + 4} = \sqrt{29}$.

- Another example
  - $\boldsymbol{x} = [5, 2], \boldsymbol{y} = [1, 4]$
    - $\boldsymbol{x} + \boldsymbol{y} = [6, 6]$
    - $||\boldsymbol{x} + \boldsymbol{y}|| = \sqrt{36 + 36} = \sqrt{72}$



3.2.2 Vector multiplication

- If c is a *scalar* and we multiply $c[x_1, x_2, ..., x_n]$, then we get $[cx_1, cx_2, ..., cx_n]$. Dividing by a scalar works the same way.

- But what about multiplying one vector by another vector? We use the **dot product**: $\boldsymbol{a} \cdot \boldsymbol{b}$. Another name for this operation is the **inner product**.

  - If $\boldsymbol{a}$ and $\boldsymbol{b}$ are both column vectors of dimension $n \times 1$, then we express the dot product as $\boldsymbol{a}^T\boldsymbol{b}$:

$$\boldsymbol{a} \cdot \boldsymbol{b} = \boldsymbol{a}^T\boldsymbol{b} = [a_1 \ \ a_2 \ \ \dots \ \ a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1b_1 + a_2b_2 + ... + a_nb_n = \sum_{i=1}^{n} a_ib_i$$

$$\text{Ex: for } \boldsymbol{a} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \text{ and } \boldsymbol{b} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \boldsymbol{a} \cdot \boldsymbol{b} = \boldsymbol{a}^T\boldsymbol{b} = [5 \ \ 2] \begin{bmatrix} 1 \\ 4 \end{bmatrix} = (5 \cdot 1) + (2 \cdot 4) = 5 + 8 = 13$$

  - Note: the result of the dot product of vectors is a *scalar*.

- The **outer product** of two vectors instead produces a matrix, written as $\boldsymbol{a}\boldsymbol{b}^T$:

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \begin{bmatrix} b_1 & b_2 \end{bmatrix} = \begin{bmatrix} a_1b_1 & a_1b_2 \\ a_2b_1 & a_2b_2 \end{bmatrix}$$

  - The dimensions of this matrix are the two outer dimensions of the vectors multiplied together:

$$\underset{3\times 1}{\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}} \underset{1\times 3}{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}} = \underset{3\times 3}{\begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}}$$

- ○ *But* the inner dimensions must match up. See 1 and 1 above. If the first matrix's number of columns is not equal to the second matrix's number of rows, then we cannot multiply.

3.3 Matrices

- A **matrix** is a rectangular table of numbers or variables arranged in a specific order in rows and columns. We express dimensions by rows, m, and columns, n. The dimensions of a matrix $\boldsymbol{A}_{m\times n}$ are pronounced 'm by n'.

$$\boldsymbol{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \in \mathbb{R}^{m\times n} = \mathbb{R}^{3\times 3}$$

- If $m = n$, then the matrix is square.

- **Types of matrices**:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \text{zero matrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \text{diagonal matrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \text{identity matrix}$$

3.4 Matrix operators

- **Addition**

  - ○ Must have the same number of elements

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \text{can't do}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & 4 \\ 6 & 8 \end{bmatrix}$$

- **Transposition**

- Rotate so that the first column becomes the first row:

$$\boldsymbol{X} = \underset{3\times 2}{\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}}, \quad \boldsymbol{X}^T = \underset{2\times 3}{\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \end{bmatrix}}$$

- **Multiplication**

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 7 & 0 & -1 \\ 1 & 3 & 1 \end{bmatrix}$$

- Because $\boldsymbol{A}$ is 2×2 and $\boldsymbol{B}$ is 2×3, we can multiply. But $\boldsymbol{AB}^T$ is undefined because $\boldsymbol{B}^T$ is 3×2.

$$\begin{aligned}
\boldsymbol{AB} &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 7 & 0 & -1 \\ 1 & 3 & 1 \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{A}_{11}\boldsymbol{B}_{11} + \boldsymbol{A}_{12}\boldsymbol{B}_{21} & \boldsymbol{A}_{11}\boldsymbol{B}_{12} + \boldsymbol{A}_{12}\boldsymbol{B}_{22} & \boldsymbol{A}_{11}\boldsymbol{B}_{13} + \boldsymbol{A}_{12}\boldsymbol{B}_{23} \\ \boldsymbol{A}_{21}\boldsymbol{B}_{11} + \boldsymbol{A}_{22}\boldsymbol{B}_{21} & \boldsymbol{A}_{21}\boldsymbol{B}_{12} + \boldsymbol{A}_{22}\boldsymbol{B}_{22} & \boldsymbol{A}_{21}\boldsymbol{B}_{13} + \boldsymbol{A}_{22}\boldsymbol{B}_{23} \end{bmatrix} \\
&= \begin{bmatrix} (1\cdot 7) + (2 \cdot 1) & (1 \cdot 0) + (2 \cdot 3) & (1 \cdot -1) + (2 \cdot 1) \\ (3 \cdot 7) + (4 \cdot 1) & (3 \cdot 0) + (4 \cdot 3) & (3 \cdot -1) + (4 \cdot 1) \end{bmatrix} = \begin{bmatrix} 9 & 6 & 1 \\ 25 & 12 & 1 \end{bmatrix}
\end{aligned}$$

- Ex with identity matrix:

$$\begin{aligned}
\boldsymbol{I}_{2\times 2}\boldsymbol{X} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\
\text{b/c} &\begin{bmatrix} (1 \cdot 1) + (0 \cdot 3) & (0 \cdot 1) + (1 \cdot 2) \\ (0 \cdot 1) + (1 \cdot 3) & (0 \cdot 3) + (1 \cdot 4) \end{bmatrix}
\end{aligned}$$

3.5 Matrix Representation of Systems of Equations

- Take this system of equations:

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m
\end{aligned}$$

- Or in matrix form:

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}}_{m\times n} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{n\times 1} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}}_{m\times 1}$$

- Ex:

$$\begin{aligned} 2x_1 + 3x_2 - 1x_3 &= 7 \\ 4x_1 + 5x_2 + 6x_3 &= 8 \\ -1x_1 + 2x_2 + 1x_3 &= 9 \end{aligned}$$

$$\Downarrow$$

$$\underbrace{\begin{bmatrix} 2 & 3 & -1 \\ 4 & 5 & 6 \\ -1 & 2 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}}_{b}$$

3.6 Linear Dependence and Independence

- The **span** of a vector is all of its linear combinations. I.e. $c \begin{bmatrix} 2 \\ 3 \end{bmatrix} \forall c$.
- If one vector falls in the span of another vector then the two are **linearly dependent**. There is no new information in the second vector.
  - $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 6 \end{bmatrix}$ are linearly dependent. The second is the first times 2.
- If one vector *does not* fall in the span of another vector, then the two are **linearly independent**.
  - $\begin{bmatrix} 7 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$ are linearly independent. One cannot be represented as a linear combination of the other.
- If we were to draw each item in the vector set, then we are interested in whether or not the overall span of a set changes when a vector is added or removed.
- In $2d$ space, if 2 vectors are linearly independent, there is some combination of those two vectors that can reach *every* point in that space. In other words, the span is the full $2d$ space. If you have 2 linearly independent vectors in $3d$ space, the span is a $2d$ plane; to reach the full space, you need a 3rd vector that is linearly independent of the other two.

Grant Sanderson has an excellent YouTube channel, 3Blue1Brown, where he describes linear (in)dependence, spans, and bases. Video.

3.7 Properties of Matrix Operators

- Matrix Addition
  - If $\boldsymbol{A}$ and $\boldsymbol{B}$ are both the same size ($m \times n$):
    1. $\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$
    2. $\boldsymbol{A} + (\boldsymbol{B} + \boldsymbol{C}) = (\boldsymbol{A} + \boldsymbol{B}) + \boldsymbol{C}$
    3. Additive Inverse: $\boldsymbol{A} + (-\boldsymbol{A}) = 0_{m \times n}$
- Matrix Multiplication:
  - If $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are conformable:
    1. $\boldsymbol{A}(\boldsymbol{BC}) = (\boldsymbol{AB})\boldsymbol{C}$
    2. $\boldsymbol{IA} = \boldsymbol{AI} = \boldsymbol{A}$
       - Where $\boldsymbol{I}$ is an identity matrix, e.g. $\boldsymbol{I}_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
    3. $\boldsymbol{A}(\boldsymbol{B} + \boldsymbol{C}) = \boldsymbol{AB} + \boldsymbol{AC} \neq \boldsymbol{BA} + \boldsymbol{CA}$ because order matters
- Matrix Exponents
  1. $\boldsymbol{A}^p \cdot \boldsymbol{A}^q = \boldsymbol{A}^{p+q}$
  2. $(\boldsymbol{A}^p)^q = \boldsymbol{A}^{pq}$
  3. $(\boldsymbol{AB})^p \neq \boldsymbol{A}^p\boldsymbol{B}^p$ unless $\boldsymbol{AB} = \boldsymbol{BA}$
- Scalar Multiplication; $\boldsymbol{A}, \boldsymbol{B}$ are matrices and r, s are scalars:
  1. $r(s\boldsymbol{A}) = (rs\boldsymbol{A})$
  2. $(r + s)\boldsymbol{A} = r\boldsymbol{A} + s\boldsymbol{A}$
  3. $r(\boldsymbol{A} + \boldsymbol{B}) = r\boldsymbol{A} + r\boldsymbol{B}$
  4. $\boldsymbol{A}(r\boldsymbol{B}) = r\boldsymbol{AB} = \boldsymbol{AB}r$
- Matrix Transposition
  1. $(\boldsymbol{A}^T)^T = \boldsymbol{A}$
  2. $(\boldsymbol{A} + \boldsymbol{B})^T = \boldsymbol{A}^T + \boldsymbol{B}^T$
  3. $(\boldsymbol{AB})^T = \boldsymbol{B}^T\boldsymbol{A}^T$
  4. $(r\boldsymbol{A})^T = r(\boldsymbol{A}^T)$
- Symmetric Matrix: A square matrix is symmetric if $\boldsymbol{A}^T = \boldsymbol{A}$
  - Ex: $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 1 & 5 \end{bmatrix}$

3.8 Idempotent Matrices

- A matrix, $\boldsymbol{A}$, is idempotent if $\boldsymbol{AA} = \boldsymbol{A}$. Multiplying the matrix by itself returns the original matrix.

3.9 Gauss-Jordan Elimination

- We can use a matrix to represent a system of equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + ... + a_{mn}x_n &= b_m \end{aligned}$$

- Rewritten as an **augmented matrix**:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array}\right]$$

- To solve for a system of equations, we often want to translate a matrix into **row echelon** or **reduced row echelon** form. The conditions are:

  1. If any rows are zeros, then they are below nonzero rows (include at least one nonzero element):
  $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 0 \end{array}\right]$
  2. The first non-zero entry in any row is 1 (except all zero row).
  $\left[\begin{array}{ccc|c} 0 & 1 & 3 & 7 \\ 0 & 0 & 0 & 0 \end{array}\right]$
  3. For each non-zero row the leading 1 is to the right of each 1 in the row above.
  $\left[\begin{array}{ccc|c} 1 & 0 & 3 & 7 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$
  4. **Reduced row-echelon form**: makes the solution to a system of equations obvious. Below $x_1 = 7, x_2 = 0, x_3 = 2$
  $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$

- Transforming a matrix into row echelon and reduced row echelon form is referred to as **Gauss-Jordan elimination**. We do so through **elementary row operators**:
    - Multiplying a row by a constant (both sides of the equation).
    - Adding/subtracting rows
    - Interchanging rows
- Example:

$$\begin{aligned} 1x_1 + 2x_2 + 4x_3 &= 3 \\ 2x_1 + 1x_2 + 3x_3 &= 2 \\ 1x_1 - 2x_2 + 2x_3 &= 3 \end{aligned}$$

    - As an augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 \\ 1 & -2 & 2 & 3 \end{array}\right]$$

    - Add $-2r1$ to $r2$ and $-1r1$ to $r3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 3 \\ 0 & -3 & -5 & -4 \\ 0 & -4 & -2 & 0 \end{array}\right]$$

    - $r3 \times -\frac{1}{4}$ and interchange $r2$ and $r3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 3 \\ 0 & 1 & 1/2 & 0 \\ 0 & -3 & -5 & -4 \end{array}\right]$$

    - Add $3r2$ to $r3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 3 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & -7/2 & -4 \end{array}\right]$$

    - Multiply $r3$ by $-\frac{2}{7}$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 3 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 1 & 8/7 \end{array}\right]$$

    - Subtract $4r3$ from $r1$ and subtract $\frac{1}{2}r3$ from $r2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -11/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & 8/7 \end{array}\right]$$

- Subtract $2r2$ from $r1$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -3/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & 8/7 \end{array}\right]$$

- $x_1 = -\frac{3}{7}, x_2 = -\frac{4}{7}, x_3 = \frac{8}{7}$
- Or: $\mathrm{I}_3 X = \begin{bmatrix} -3/7 \\ -4/7 \\ 8/7 \end{bmatrix}$

3.10 Matrix Inversion

- For a scalar a, the inverse: $a^{-1} = \frac{1}{a}$, where $a^{-1}a = 1$.
- We can also invert a matrix A: $A^{-1}A = AA^{-1} = \mathrm{I}$
  - Note: if A^{-1} does not exist, then A is singular/not invertible.
  - $A^T \neq A^{-1}$
- For a 2×2 matrix, we use the following steps:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Where $ad - bc$ is the *determinant*, which we discuss more later. An example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$
$$A^{-1} = \frac{1}{(1 \cdot 4) - (3 \cdot 2)} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$
$$= \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

- How to check if $A^{-1}A = \mathrm{I}$?

$$\begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

```r
# To check this in R:
> (A <- rbind(c(1, 2), c(3, 4)))
> (Ainv <- solve(A))
> A %*% Ainv
```

3.11 Properties of Matrix Inversion

Singular vs nonsingular:

- Invertible = Nonsingular
- Noninvertible = Singular
- If A is nonsingular, then A^{-1} is nonsingular
- If A & B are nonsingular, then $(AB)^{-1} = B^{-1}A^{-1}$
- If A is nonsingular, then $(A^T)^{-1} = (A^{-1})^T$

Some conditions for nonsingularity:

1. Rows and columns are linearly independent (the matrix has full rank).
2. Matrix A is row equivalent to I.
3. The determinant is not 0.

One intuitive and practical procedure for finding A^{-1}, regardless of the size:

1. Find $[A \mid \mathrm{I}]$ where both A and I are both $n \times n$
2. Find the reduced row echelon form for A (left side)
3. If step 2 gives us $[\mathrm{I} \mid C]$, then $C = A^{-1}$

A note on rank: If A is $m \times n$, then the rank(A) is the maximum number of linearly independent rows or columns in the matrix.

- E.g.: $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{1} & 2 & 3 \\ \mathbf{2} & 4 & 6 \end{bmatrix}$ Only the bold rows and columns are linearly ind. So, $\text{rank}(A) = 1$

- **Examples:**

$$A\mathrm{I} = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & 0 & 1 & 0 \\ 5 & 5 & 1 & 0 & 0 & 1 \end{array}\right]$$

- Subtract 1/2 r2 from r1, divide r2 by 2, subtract 5r1 from r3

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & -1/2 & 1 & -1/2 & 0 \\ 0 & 1 & 3/2 & 0 & 1/2 & 0 \\ 0 & 0 & -4 & -5 & 0 & 1 \end{array}\right]$$

- Subtract 1/8 r3 from r1, add 3/8 r3 to r2, divide r3 by -4

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 13/8 & -1/2 & -1/8 \\ 0 & 1 & 0 & -15/8 & 1/2 & 3/8 \\ 0 & 0 & 1 & 5/4 & 0 & -1/4 \end{array}\right]$$

- Where the right-hand side matrix is A^{-1}

- Another example, but A^{-1} doesn't exist:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 1 & 0 \\ 5 & -2 & -3 & 0 & 0 & 1 \end{array}\right]$$

- Subtract r1 from r2, subtract 5r1 from r3

$$\begin{bmatrix} 1 & 2 & 3 & \dots & \dots & \dots \\ 0 & -4 & 4 & \dots & \dots & \dots \\ 0 & -12 & 12 & \dots & \dots & \dots \end{bmatrix}$$

- We aren't concerned with the right-hand side matrix because the rows of the left matrix are not independent. r3 is a linear combination of r2, meaning the matrix is singular (not invertible).

3.12 Determinant

- Determinants convert a square matrix into a scalar. Determinants are useful for checking if a matrix is invertible. The formula is straightforward for a 2×2 matrix, but less so for larger matrices.

- Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
  - The determinant of A is:
    $|A| = (a_{11} \cdot a_{22}) - (a_{21} \cdot a_{12})$

- Examples:
  - $\begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} \Rightarrow (2 \cdot 5) - (3 \cdot 1) = 7$

- $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \Rightarrow 2 - 2 = 0$

- For a 3×3 matrix we sum the products of all elements in any row or column, alternating signs, and the determinants of a specific 2×2 submatrix. An element's submatrix is the remaining elements when the elements from the relevant row and column are removed. I.e., for:

  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

  The submatrix for a_{23} is $\begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}$

  Taking the first column, the determinant of A is:

  $a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$

- We can use any row or column. But it is best to use one with zeros, if available.

- The **minor** of an element is the determinant of its submatrix:

  - $M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = (a_{21} \cdot a_{33}) - (a_{31} \cdot a_{23})$

3.12.1 Determinants via Cofactor Expansion

- The **cofactor** of any element i, j: $C_{ij} = (-1)^{i+j} M_{ij}$, which is used for calculating the determinants of $n \times n$ matrices where $n > 2$. i is rows, j is columns.

  - $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$
  - Ex: $C_{11} = (-1)^{1+1}\, M_{11} = 1 \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

- The determinant of a $n \times n$ matrix where $n > 2$ is the sum of the products of each element and its cofactor for any row or column. We just choose a single row or a single column – ideally one with as many zeros as possible.

  - Given row i:

    $$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij}$$

  - Or column j:

    $$\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij}$$

Let's show what this means:

- Choose the row or column with the most zeros:
  - Ex: $\begin{bmatrix} 1 & 2 & -3 & 4 \\ -4 & 2 & 1 & 3 \\ 0 & 0 & 0 & -3 \\ 2 & 0 & -2 & 3 \end{bmatrix}$
  - Let's take row 3:

$$
\begin{aligned}
|A| = \sum_{j=1}^{4} a_{3j}C_{3j} &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} + a_{34}C_{34} \\
a_{34}C_{34} &= 0 + 0 + 0 + a_{34}C_{34} \\
a_{34}C_{34} &= -3\left[(-1)^{3+4}\begin{vmatrix} 1 & 2 & -3 \\ -4 & 2 & 1 \\ 2 & 0 & 2 \end{vmatrix}\right] \\
&= -3 \cdot -1\left[2\begin{vmatrix} 2 & -3 \\ 2 & 1 \end{vmatrix} - 0 + 2\begin{vmatrix} 1 & 2 \\ -4 & 2 \end{vmatrix}\right] \\
&= 3\left[2(2+6) + 2(2+8)\right] \\
&= 3[16 + 20] = 3(36) = 108
\end{aligned}
$$

- To wrap up, some useful properties of determinants:

1. $|A| = |A^T|$
2. If a row or column of A is a linear combination of other rows or columns, then $\det(A) = 0$
3. If A is diagonal, then $|A|$ is the product of the diagonals.
4. $|AB| = |A| \cdot |B|$
5. If A is non-singular, then $|A^{-1}| = \frac{1}{|A|}$

3.13 Adjoint Matrix

- **Adjoint matrix** (also called the "adjugate"): the transpose of a matrix of the cofactors of each element:

$$
\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & \dots & C_{1n} \\ C_{21} & \dots & \dots & C_{2n} \\ \vdots & & & \\ C_{n1} & \dots & \dots & C_{nn} \end{bmatrix}^T
$$

Where C_{ij} is an element's cofactor. For example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow C = \begin{bmatrix} +\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} & -\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} & +\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ \\ -\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} & +\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} \\ \\ +\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} & +\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}^T$$

Where the last matrix is the adjoint matrix. This gives us another way to calculate the inverse of A:

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

3.14 Trace

The **trace** of a $n \times n$ matrix is the sum of its diagonal elements:

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + ... + a_{nn}$$

4 Calculus: Limits, Differentiation, and Optimization

Before we begin introducing mathematical formulas, what is a derivative? A derivative calculates the rate of change of a function at any given point. In high school when we learned about the slope of a line — $y = mx + b$, where m is the slope — that slope is a derivative. In that context, the derivative is the same at all points because the line is straight. But we often encounter functions that are not a straight line and we care about calculating the slope across values of that function. We can use calculus for this.

4.1 Sequences and Limits

A sequence is an ordered list of numbers, e.g.

$$\{x_n\} = \{x_1, x_2, ..., x_n\}$$

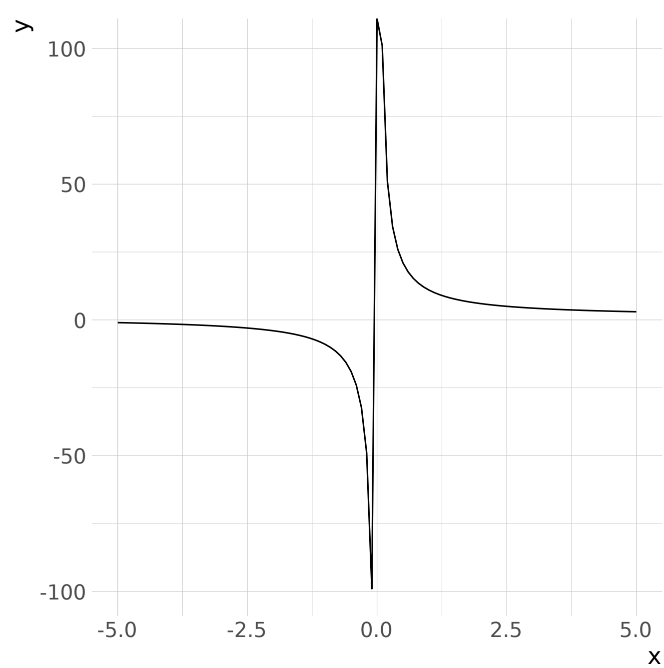
Where x_n is a real number that extends from x_1 to x_n. We usually encounter n extending to ∞. Another way to write the series is:

$$\{x_n\}_{n=1}^{\infty}$$

Central to calculus is the notion of a sequence "converging to a limit", generally where $n \to \infty$ or $n \to 0$. This is written as:

$$\lim_{n\to\infty} y_n = L$$

where L is the limit. Let's visualize for an arbitrary function, $f(x) = \frac{x+10}{x}$:



We can see that as $x \to \infty$ the value of $f(x)$ stabilizes. Indeed:

$$\lim_{x\to\infty} \frac{x+10}{x} = \lim_{x\to\infty}\left(1+\frac{10}{x}\right) = \underbrace{\lim_{x\to\infty} 1}_{1} + \underbrace{\lim_{x\to\infty} \frac{10}{x}}_{0} = 1$$

This gives us a limit of 1.

4.2 Derivatives and the Difference Quotient

The derivative is the rate of change of $f(x)$ at any x. For a straight line, i.e $y = mx + b$, the derivative is constant at all points. But for a nonlinear function, i.e. $y = 2x^4$, the rate of change varies across x.

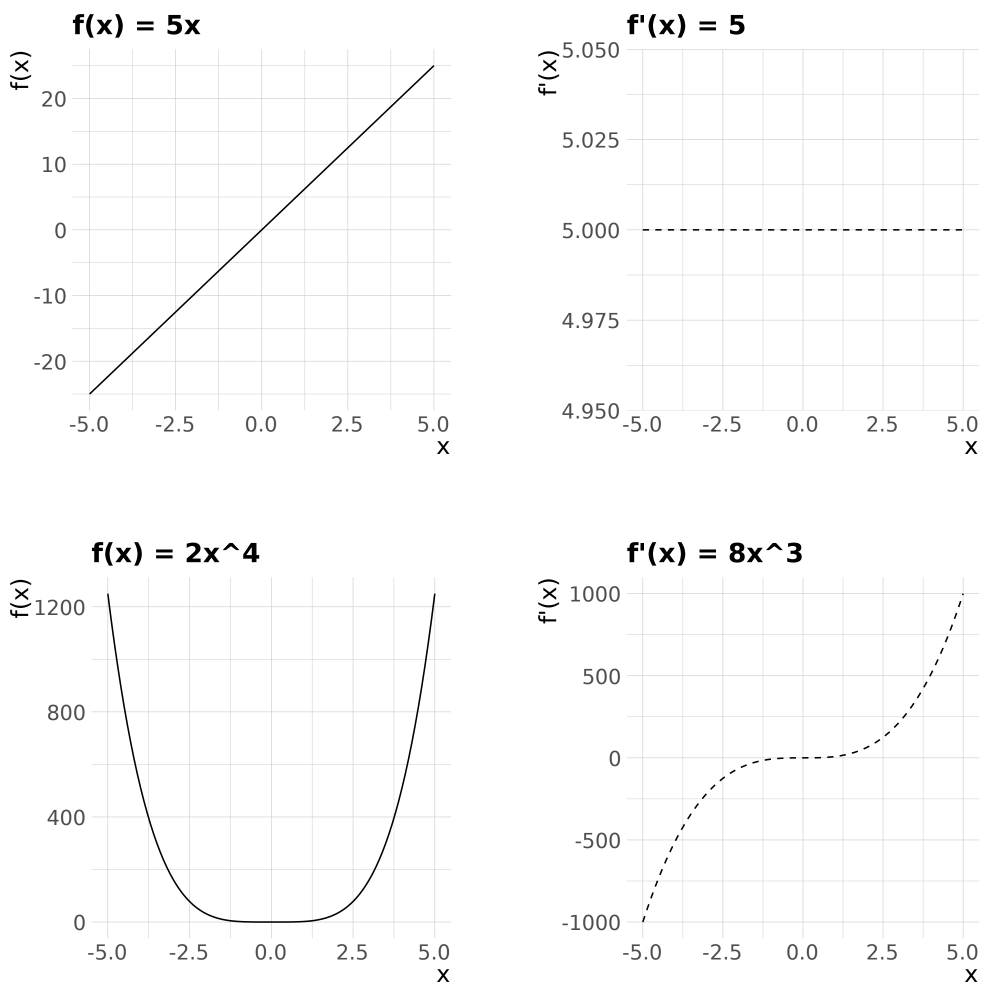


Figure 1: Functions and their derivatives

Now, let's introduce how we produce a derivative. Let f be a function with an open interval that contains x. Let h be the interval where $f(x)$ changes. Below we are simply calculating rise over run at each specified interval.

$$\frac{d}{dx}f(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The numerator represents the change in $f(x)$ as Δx approaches 0 and the denominator is the change in Δx – or the change in x. We generally see two notations for derivatives:

- Lagrange: $f'(x)$
- Leibniz: $\frac{d}{dx}y = \frac{dy}{dx}$
  - The change in y, given change in x.

Side-note, on the problem set, when we are asked about the difference quotient as a function of $x + \Delta x$, think about how a specified function would look if it were plugged in for $f(x)$ and $f(x + \Delta x)$.

4.3 Rules for Derivatives

- Power rule:
  - $y = f(x) = ax^n$, $f'(x) = nax^{n-1}$
- Constant multiplier rule:
  - $f(x) = ax$, $f'(x) = a$
- Constant rule:
  - $f(x) = a$, $f'(x) = 0$
- Summation rule:
  - $f(x) = g(x) + h(x)$, $f'(x) = g'(x) + h'(x)$
- Product rule:
  - $f(x) = g(x)h(x)$
  - $f'(x) = g'(x)h(x) + g(x)h'(x)$
- Quotient rule:
  - $f(x) = g(x)/h(x)$
  - $f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{[h(x)]^2}$
- Chain rule:
  - $f(x) = h[g(x)]$
  - $f'(x) = h'[g(x)] \cdot g'(x)$
- Exponent rule:
  - $f(x) = a^x$

- $f'(x) = a^x(ln(a))$
- $f(x) = e^x$, $f'(x) = e^x$

- Logarithm rule:
  - $f(x) = \log_a x$
  - $f'(x) = \frac{1}{x \ln a}$
  - Natural log: $f(x) = ln(x)$, $f'(x) = \frac{1}{x}$

4.4 Derivative Examples

1. $f(x) = 40x^{400}$
   - $f'(x) = 400 \cdot 40x^{399} = 16000x^{399}$
2. $f(x) = 16x$
   - $f'(x) = 16$
3. $f(x) = 1000$
   - $f'(x) = 0$
4. $f(x) = 3x^{100} + 5x^2$
   - $f'(x) = 300x^{99} + 10x$
5. $f(x) = (3x+5)(9x+2)$
   - $f'(x) = 3(9x+2) + 9(3x+5)$
6. $f(x) = \frac{3x+5}{9x+2}$
   - $f'(x) = \frac{3(9x+2) - 9(3x+5)}{(9x+2)^2}$
7. $f(x) = 20(x+3)^{10}$
   - $f'(x) = 200(x+3)^9 \cdot 1$
8. $f(x) = 20(x^3 + 3x)^{10}$
   - $f'(x) = 200(x^3 + 3x)^9 \cdot (3x^2 + 3)$
9. $f(x) = 20[(x+3)^3 + 4x]^{10}$
   - $f'(x) = 200[(x+3)^3 + 4x]^9 \cdot g'(x)$
   - $g'(x) = 3(x+3)^2 \cdot 1 + 4$
   - $f'(x) = 200[(x+3)^3 + 4x]^9 \cdot 3(x+3)^2 + 4$

10. $f(x) = e^{\sqrt{x}}$, $f'(x) = e^{x^{\frac{1}{2}}} \cdot (\frac{1}{2}x^{-\frac{1}{2}})$

11. $f(x) = a^{\sqrt{x}}$, $f'(x) = a^{\sqrt{x}} ln(a) \cdot (\frac{1}{2}x^{-\frac{1}{2}})$

    - Keep in mind for the problem set.

12. $f(x) = ln(3x^2 + 3x + 3)$, $f'(x) = \frac{1}{3x^2+3x+3} \cdot 6x + 3 = \frac{6x+3}{3x^2+3x+3}$

4.5 More on the Chain Rule

Let's revisit the chain rule in more detail. Another way to write it out is:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

So we first specify one of the functions as u and then calculate $\frac{dy}{du}$. Then we calculate $\frac{du}{dx}$, which is the change in the function u, given a change in x. Last, we write out the product of the two derivatives. This is an incredibly common and powerful trick. One more example:

$$\begin{aligned}
y &= \left[\frac{(x^2+1)(3x+9)}{(8x^2-9)}\right]^3 \\
u &= \frac{(x^2+1)(3x+9)}{(8x^2-9)} \\
\frac{dx}{dy} &= 3[u]^2 \cdot u'
\end{aligned}$$

Where we solve for u' with the product and quotient rules.

4.6 Implicit Differentiation

Sometimes we cannot fully isolate x and y when finding $\frac{dy}{dx}$. This means we cannot solve for y solely in terms of x. We use implicit differentiation for these situations. You'll notice that our answer to the following example includes both x and y. The trick is to write out all parts of our function in terms of $\frac{dy}{dx}$. Then we solve for $\frac{dy}{dx}$.

$$\begin{aligned}
&f(x,y) = y^5 + xy + x^2 \quad \text{at} \quad f(x,y) = 3 \\
&\frac{d}{dx}f(x,y) = \frac{d}{dx}(3) \\
&\frac{d}{dx}(y^5 + xy + x^2) = \frac{d}{dx}(3) \\
&5y^4\frac{dy}{dx} + (y + x\frac{dy}{dx}) + 2x = 0 \\
&(5y^4 + x)\frac{dy}{dx} = -y - 2x \\
&\frac{dy}{dx} = \frac{-y-2x}{5y^4+x}
\end{aligned}$$

Now, we have a quantity for $\frac{dy}{dx}$, it just depends upon the values of x *and* y.

4.7 Partial Derivatives

What if we have a multivariate function (multiple variables are changing) and are only concerned with the change in y, given the change in *one* variable? The trick is to treat other variables as a constant, because we are only concerned with variation in one variable. So:

$$\begin{aligned} f(x,y) &= 5x^2y^3 \\ \frac{df}{dx} &= 10xy^3 \\ \frac{df}{dy} &= 15x^2y^2 \end{aligned}$$

Some additional notation: $\frac{df}{dx} = f_x$ or $\frac{df}{dy} = f_y$
Examples:

$$\begin{aligned} f(x,y) &= (x+4)(3x+2y) \\ f_x &= 1\cdot(3x+2y)+(x+4)(3) \\ f_y &= 0(3x+2y)+(x+4)2 \\ &= 2(x+4) \end{aligned}$$

But:

$$\begin{aligned} f(x,y,z) &= (x+4)(3x+2y)(3z) \\ f_z &= 3(x+4)(3x+2y) \end{aligned}$$

This last example is simpler because the other parts of the function don't include z, so we treat them like constants and then derive $3z$.

4.8 Gradient

A vector with all of a function's partial derivatives is the **gradient**. Take a function $f(x)$, with n variables. The gradient – or $\nabla f(x)$ – is:

$$\nabla f(x) = \left[\frac{df(x)}{dx_1}, \frac{df(x)}{dx_2}, \ldots, \frac{df(x)}{dx_n}\right]$$

For a function $f(x,y) = \left[\frac{df}{dx}, \frac{df}{dy}\right] = [f_x, f_y]$

4.9 Second Derivatives and the Hessian

We can take higher-order derivatives, which are just the rate of change of the previous derivative. I.e. $f(x) = x^3$, $f'(x) = 3x^2$, $f''(x) = 6x$, and so on...
In your statistics courses you will encounter the **Hessian**, which is a matrix of all combinations of second derivatives:

$$\begin{bmatrix} \frac{d^2f}{dx_1dx_1} & \cdots & \frac{d^2f}{dx_ndx_1} \\ \frac{d^2f}{dx_2dx_1} & \frac{d^2f}{dx_2dx_2} & \vdots \\ \vdots & & \ddots \\ \frac{d^2f}{dx_ndx_1} & \cdots & \frac{d^2f}{dx_ndx_n} \end{bmatrix}$$

This gives us the change of the curvature of a function in all directions.

4.10 More on Limits

We can also take 'one-sided' limits, which come from different directions. Consider $f(x) = \frac{1}{x}$:

$$\lim_{x \to 0^+} \frac{1}{x} = +\infty$$
$$\lim_{x \to 0^-} \frac{1}{x} = -\infty$$

We also sometimes need to simplify a function to find the limit. In this example, plugging 1 in gives us 0/0. But simplifying gives us 2.

$$\lim_{x \to 1} \frac{1-x^2}{1-x} = \frac{(1-x)(1+x)}{1-x} = 1 + x = 2$$

4.11 L'Hôpital's Rule

One more trick for limits: If $\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0$ or $= \pm\infty$, *and* $\lim_{x\to c} \frac{f'(x)}{g'(x)}$ exists, then:

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}$$

This is useful for undefined limits. Ex:

$$
\begin{aligned}
\lim_{x \to 1} \frac{x^2 - 1}{x^2 + 3x - 4} &= \frac{1^2 - 1}{1^2 + 3(1) - 4} = \frac{0}{0} \\
&= \lim_{x \to 1} \frac{2x}{2x + 3} \\
&= \frac{2(1)}{2(1) + 3} \\
&= \frac{2}{5}
\end{aligned}
$$

4.12 Optimization

We often want to know where our functions are at their (local) maximum or minimum. We do this in two steps. First, wherever the first derivative is at 0, means we are either at a minimum or maximum. Second, depending upon the direction of the second derivative, we can tell if we are at a maximum or minimum.

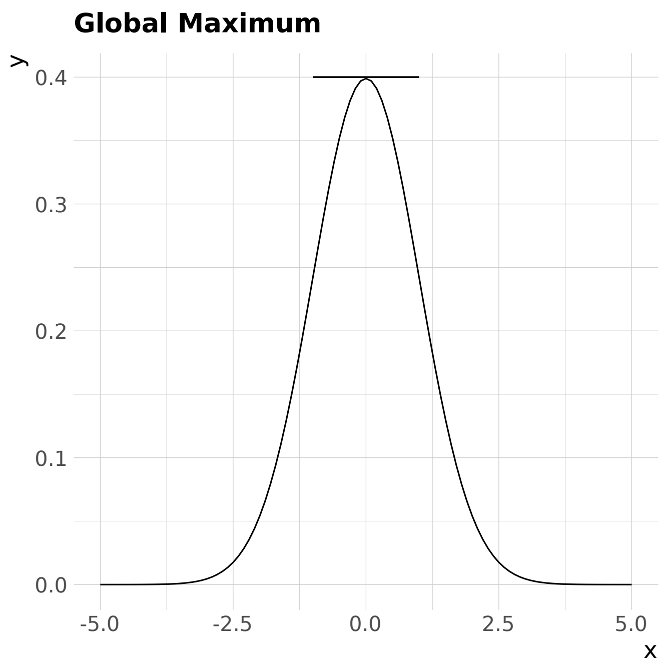


Figure 2: Visualizing how a derivative is equal to 0 at a function's maximum or minimum.

- **First-order condition**: $f'(x) = 0$
  - Maximum or minimum
  - This assumes that $f(x)$ is differentiable at all values; i.e., it's a continuous function.
- **Second-order condition**: $f''(x)$, where > 0 means minimum, < 0 means maximum, and $= 0$ an inflection point.

$$
\begin{aligned}
f(x) &= x^2 - 4x - 1 \\
\text{FOC: } f'(x) &= 2x - 4 \\
x &= 2 \\
\text{SOC: } f''(x) &= 2 \Rightarrow \text{minimum}
\end{aligned}
$$

What if we have multiple variables?

- Let $f(x) = -x_1^2 + x_1 x_2 - x_2^2$
    - FOC:
        - $f_{x_1} = -2x_1 + x_2 = 0$
        - $f_{x_2} = x_1 - 2x_2 = 0$

        $$\left[\begin{array}{cc|c} -2 & 1 & 0 \\ 1 & -2 & 0 \end{array}\right]$$

            - It turns out $x_1 = 0$, $x_2 = 0$
    - SOC: We need to build the Hessian:

    $$\begin{bmatrix} \frac{d^2 f}{dx_1 x_1} & \frac{d^2}{dx_1 x_2} \\ \frac{d^2 f}{dx_2 x_1} & \frac{d^2 f}{dx_2 x_2} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

- Now, we ask whether these points are minima, maxima, indeterminate, or saddle points.
    - We calculate the determinants (D_i) of each "principal minors" (PM_i).
    - $\text{PM}_1 = -2$, $D_1 = -2$
    - $\text{PM}_2 = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$, $D_2 = (-2 \cdot -2) - (1 \cdot 1) = 3$
- The Hessian is:
    - Positive definite: $D_1 > 0$, $D_2 > 0$, strictly local minima ["concave" or]
    - Negative definite: $D_1 < 0$, $D_2 > 0$, strictly local maxima
        - Start negative and then alternate signs
    - Positive semi-definite: $D_1 \geqslant 0$, $D_2 \geqslant 0$
    - Negative semi-definite: $D_1 \leqslant 0$, $D_2 \geqslant 0$
- For larger matrices?
    - $\begin{bmatrix} . \end{bmatrix}$, $\text{PM}_1 = \text{determinant of } 1 \times 1$
    - $\begin{bmatrix} . & . \\ . & . \end{bmatrix}$, $\text{PM}_2 = \text{determinant of } 2 \times 2$
    - $\begin{bmatrix} . & . & . \\ . & . & . \\ . & . & . \end{bmatrix}$, $\text{PM}_3 = \text{determinant of } 3 \times 3$

- If we have a 3×3 matrix:
  - PD: $\mathrm{D}_1 > 0$, $\mathrm{D}_2 > 0$, $\mathrm{D}_3 > 0$
  - ND: $\mathrm{D}_1 < 0$, $\mathrm{D}_2 > 0$, $\mathrm{D}_3 < 0$
  - PSD: $\mathrm{D}_1 \geqslant 0$, $\mathrm{D}_2 \geqslant 0$, $\mathrm{D}_3 \geqslant 0$
  - NSD: $\mathrm{D}_1 \leqslant 0$, $\mathrm{D}_2 \geqslant 0$, $\mathrm{D}_3 \leqslant 0$

4.13 Constrained Optimization – Lagrange Multiplier

Sometimes, we have to find the maxima or minima under set conditions.

$$\text{Max } f(x_1, x_2) \text{ s.t. } g(x_1, x_2) = 0$$
$$\text{Max } f(x_1, x_2) - \lambda g(x_1, x_2)$$

- λ is us creating a new variable.
- FOC:

$$\begin{aligned} f_{x_1} - \lambda g_{x_1} &= 0 \\ f_{x_2} - \lambda g_{x_1} &= 0 \\ g(x_1, x_2) &= 0 \end{aligned}$$

- Then:

$$\begin{aligned} f(x_1, x_2) =& 36 - x_1^2 - x_2^2, \quad g(x_1, x_2) = x_1 + 7x_2 - 25 \\ f(x_1, x_2, \lambda) =& 36 - x_1^2 - x_2^2 - \lambda(x_1 + 7x_2 - 25) \\ f_{x_1} &= -2x_1 - \lambda = 0, \quad \Rightarrow \quad x_1 = -\lambda/2 \\ f_{x_2} &= -2x_2 - 7\lambda = 0, \quad \Rightarrow \quad x_2 = -7\lambda/2 \\ g(x_1, x_2) &= -\lambda/2 + -7(7\lambda/2) - 25 = 0 \\ \lambda &= -1 \\ \text{Therefore: } x_1 &= \frac{1}{2},\ x_2 = \frac{7}{2} \end{aligned}$$

5 Calculus: Integration

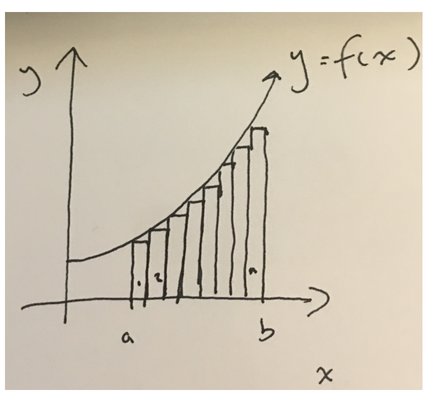
We integrate so that we can add continuously. The integral is the limit of the sum, where we add all slices under the curve until we reach infinity. We end up summing the values of infinitesimally small ranges. This is the idea behind Riemann sums.

5.1 Indefinite Integrals and Antidifferentiation

- $f(x) = 40x$
- $F(x) = 20x^2 + c$
  - This is the 'antiderivative', $F'(x) = f(x)$, because if $F(X)$ is derived then it returns $f(x)$.
  - We add c because any constant will turn to 0 when the derivative is taken.
- Powerfully, antidifferentiation lets us solve for the value of y, given dy/dx.
- **Rules of integration**
  - Power rule: $f(x) = x^n$, $F(x) = \frac{1}{n+1}x^{n+1} + c$
  - Exponent rule: $f(x) = e^x$, $F(x) = e^x + c$
    - But: $f(x) = e^{ax}$, $F(x) = \frac{1}{a}e^{ax} + c$
  - Logarithm rule: $f(x) = \frac{1}{x}$, $F(x) = ln(x) + c$
  - Chain rule: $f(x) = g'(x)e^{g(x)}$, $F(x) = e^{g(x)} + c$
  - Chain + log: $f(x) = \frac{g'(x)}{g(x)}$, $F(x) = ln|g(x)| + c$
  - Sum rule: $f(x) = g(x) + h(x)$, $F(x) = G(x) + H(x)$
  - Constant rule: $f(x) = kg(x)$, $F(x) = kG(x)$
  - Integral of a constant: $f(x) = k$, $F(x) = kx + c$

5.2 Areas and Riemann Sums

Consider the following function:



where $\sum_{i=1}^{n} f(x_i)\Delta x$ and $\Delta x = \frac{b-a}{n}$. This equals the sum of the area of all the rectangles under $f(x)$. The closer n gets to ∞, then the closer the sum of all the areas gets to the area under $f(x)$ from a to b.

It also turns out that:

$$\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\Delta x = \int_a^b f(x)dx$$

Because of this, we can use a definite integral to calculate the area under a function/curve between two points.

5.3 Fundamental Theorem of Calculus

This brings us to the grandiose titled fundamental theorem of calculus:

$$\int_a^b f(x)dx = F(b) - F(a)$$

Where the difference between the antiderivative at two points gives us the area under a curve between these two values. This is given such importance because it links derivatives and integrals.
Ex:

$$\int_1^3 x^2 dx = \frac{1}{3}x^3\Big|_1^3 = \frac{27}{3} - \frac{1}{3} = \frac{26}{3}$$

5.4 Rules for Definite Integrals

1. $$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

2. $$\int_a^d f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx + \int_c^d f(x)dx \quad \text{where} \quad a \leqslant b \leqslant c \leqslant d$$

3. $$\int_a^b -f(x)dx = -\int_a^b f(x)dx$$

4. $$\int_a^b kf(x)dx = k\int_a^b f(x)dx$$

5. $$\int_a^b [f(x) \pm g(x)]\, dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$$

5.5 Integration by Substitution

1. Define a new variable, $u = G(x)$ such that when written in terms of u the integrand is simpler.

2. Integrate the function, changing the limits of the integral to reflect u.

3. Rewrite in terms of x.

Ex:

$$\int_{x=a}^{x=b} 8x(x^2+1)^3 dx$$

- Let $u = x^2 + 1$
- Then, $\frac{du}{dx} = 2x$
- And: $\frac{1}{2x} du = dx$. So we can rewrite the above function as:

$$\int_a^b 8xu^3 \cdot \frac{1}{2x} du$$

We simplify to change the limits to reflect $g(u)$.

$$\int_{a^2+1}^{b^2+1} 4u^3 du$$
$$= \int_{a^2+1}^{b^2+1} \frac{4}{4} u^4 = u^4 \Big|_{a^2+1}^{b^2+1}$$

Then we rewrite in terms of x:

$$(x^2+1)^4 \Big|_a^b$$

5.5.1 Some examples

-
$$
\begin{aligned}
&\int_a^b 3x^2\sqrt{x^3+1}dx \\
&\text{Let } u = x^3+1 \\
&\frac{du}{dx} = 3x^2,\ \frac{1}{3x^2}du = dx \\
&= \int_{a^3+1}^{b^3+1} 3x^2\sqrt{u}\frac{1}{3x^2}du \\
&= \int_{a^3+1}^{b^3+1} \sqrt{u}\,du = \int_{a^3+1}^{b^3+1} u^{\frac{1}{2}}du \\
&= \frac{2}{3}u^{\frac{3}{2}}\Big|_{a^3+1}^{b^3+1} = \frac{2}{3}(x^3+1)^{\frac{3}{2}}\Big|_a^b
\end{aligned}
$$

-
$$
\begin{aligned}
&\int_a^b x^2e^{x^3}dx \\
&\text{Let } u = x^3, \frac{du}{dx} = 3x^2, dx = \frac{du}{3x^2} \\
&\int_{a^3}^{b^3} x^2e^u\frac{du}{3x^2} = \frac{1}{3}\int_{a^3}^{b^3} e^u du \\
&= \frac{1}{3}e^u\Big|_{a^3}^{b^3} = \frac{1}{3}e^{x^3}\Big|_a^b
\end{aligned}
$$

-
$$
\begin{aligned}
&\int_a^b \frac{2-x}{\sqrt{2x^2-8x+1}}dx \\
&\text{Let } u = 2x^2-8x+1 \\
&\frac{du}{dx} = 4x-8, dx = \frac{du}{4x-8} \\
&\int (2-x)u^{-\frac{1}{2}}\frac{du}{-4(2-x)} = -\frac{1}{4}\int u^{-\frac{1}{2}}du \\
&= -\frac{1}{4}\cdot\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \\
&= -\frac{1}{2}u^{\frac{1}{2}}\Big|_a^b
\end{aligned}
$$

5.6 Integration by Parts

Sometimes we get functions where integrating by substitution does not gain us any leverage. For these, we can *integrate by parts*. Say we have two functions u and v, which are differentiable. We can use the product rule to show that:

$$\frac{d}{dx}(uv) = v\frac{du}{dx} + u\frac{dv}{dx}$$

And rearrange:

$$u\frac{dv}{dx} = \frac{d}{dx}(uv) - v\frac{du}{dx}$$

Integrate with respect to x, and we get the formula for integration by parts:

$$\int udv = uv - \int vdu$$

An example:

$$\begin{aligned}
&\int_a^b x^2 \ln(x)\,dx \\
&u = \ln(x) \Rightarrow du = \frac{1}{x}\,dx \\
&dv = x^2 dx \quad \text{Integrate } \Rightarrow v = \frac{x^3}{3} \\
&\text{So the original equation equals:} \\
&\frac{1}{3}x^3 \ln x - \int \frac{1}{3}x^3 \cdot \frac{1}{x} dx \\
&= \frac{1}{3}x^3 \ln x - \frac{1}{3}\int x^2 dx = \left[\frac{1}{3}x^3 \ln x - \frac{1}{3}\cdot\frac{1}{3}x^3\right]_a^b
\end{aligned}$$

One more:

$$\begin{aligned}
&\int_1^3 xe^x dx \\
&u = x, du = dx, dv = e^x dx, v = e^x \\
&\int_1^3 xe^x dx = xe^x - \int e^x dx \\
&= [xe^x - e^x]_1^3
\end{aligned}$$

5.7 Improper Integrals

What if we integrate to ∞ instead of an integer? We take the limit when we get to plugging our bounds in.

$$
\begin{aligned}
&\int_1^\infty \frac{1}{\lambda} e^{-\lambda x} dx \\
&= \frac{1}{\lambda} \int_1^\infty -\frac{1}{\lambda} e^{-\lambda x} = \left[-\frac{1}{\lambda^2} e^{-\lambda x} \right]_1^\infty \\
&= \left[\lim_{x\to\infty} -\frac{1}{\lambda^2} e^{-\lambda x} \right] - \left[-\frac{1}{\lambda^2} e^{-\lambda 1} \right] \\
&= 0 + \left[\frac{1}{\lambda^2} e^{-\lambda} \right]
\end{aligned}
$$

5.8 Double Integrals

Sometimes we want to integrate the area of a function depending on multiple variables. In this case we can take a double integral. We just take the integral of the function with respect to one variable, and then we integrate the outcome with respect to another variable:

$$
\begin{aligned}
&\int_{x_2=0}^1 \int_{x_1=-1}^1 (x_1^2 + x_2^2) dx_1 dx_2 \\
=& \int_0^1 \left[\int_{-1}^1 (x_1^2 + x_2^2) dx_1 \right] dx_2 \\
=& \int_0^1 \left[\frac{1}{3} x_1^3 + x_1 x_2^2 \right]_{-1}^1 dx_2 \\
=& \int_0^1 \left\{ \left[\frac{1}{3} 1^3 + 1 x_2^2 \right] - \left[\frac{1}{3} (-1)^3 + (-1) x_2^2 \right] \right\} dx_2 \\
=& \int_0^1 \left(\frac{2}{3} + 2 x_2^2 \right) dx_2 \\
=& \left[\frac{2}{3} x_2 + \frac{2}{3} x_2^3 \right]_0^1 \\
=& \frac{4}{3}
\end{aligned}
$$

References

W. H. Moore and D. A. Siegel. *A mathematics course for political and social research*. Princeton University Press, 2013.